

Carl Frederick Valone

Technical Lead, Senior UI Engineer, Front-end Architect

Louisburg, NC 27549 | +1 919 340 3119 | freddie.valone@gmail.com

Professional Summary

Accomplished Senior Technical Lead and Front-end Architect with 15+ years of experience in web application development and UI development. Expertise in modern frontend frameworks such as Vue.js, React, react-router v7, Nx Monorepo, Bootstrap, TailwindCSS, and AngularJS, with a proven track record of leading teams to deliver high-quality projects on time. Skilled in mentoring junior developers and implementing responsive design features, enhancing user experience, and optimizing application performance. Committed to leveraging advanced technical knowledge to drive innovation in front-end development.

Work Experience

Senior Technical Lead, UI Engineer, Front-end Architect

U. S. Environmental Protection Agency | Durham, NC

July 2018 to Present

Senior Technical Lead UI Front-end Architect

- Determine UI architectural stack including VueJS, NuxtJs bootstrap, TailwindCSS, ag-grid; for tabular data visualization, and Cypress UI testing.
- Detailed analysis of the legacy web applications (RoR, MySQL, GraphQL VueJs hybrid) for conversion to JAVA REST, NUXT/VueJS, Postress/Mongo DB applications.
- Create and maintain a Bitbucket (GIT) repository of Architectural Decision Records for the new application UI.
- Maintain linting standards and document proper installation of linting tools and configuration for multiple IDEs.
- Create new EPA Center for Computational Toxicology and Exposure UI base application template as a Bitbucket repository.
- Lead team to migrate our Nuxt 2/Vue 2 Bootstrap applications; Rapidtox, GenRA (Generalized Read Across) , and Comptox Chemicals Dashboard (ongoing) to Nuxt 3/Vue 3 with Composition API and typescript. Integrating PrimeVue component library, TailwindCSS, and Vitest component/unit testing
- Provide architecture and serve as UI lead for the creation of a separate application template repository for the EPA CTE web applications.
- Update application framework as needed for security patches.
- Continually stay abreast of developments and advances in the JavaScript, HTML, CSS and web development areas and apply sensible changes to the application architecture as new technology might dictate.
- Provided guidance and mentorship to junior UI developers on design processes, tools, coding best practices, providing guidance on industry trends and emerging technologies.
- Manage a team of front-end developers in the successful completion of multiple web development projects, ensuring adherence to project timelines and quality standards.
- Conduct code reviews to ensure adherence to coding standards, maintainability, scalability, and security measures.
- Worked closely with back-end developers to integrate front-end components with server-side logic using RESTful APIs or other methods.
- Served as a technical lead in client meetings or presentations discussing project requirements, progress updates, and technical feasibility.
- Collaborate with cross-functional teams including scientists, Data Engineers, JAVA developers, DBA, Product Owners, Product Managers, QA testers, and stakeholders to gather requirements and define project goals throughout the project lifecycle.
- Pioneered the adoption of new tools or technologies within the team that improved efficiency, developer experience, and enhanced user experience on websites/applications.
- Participate in Agile/Scrum ceremonies such as daily stand-ups, sprint planning, backlog grooming, and retrospective meetings.

Example URL

- <https://comptox.epa.gov/dashboard>
- <https://comptox.epa.gov/genra/>
- <https://comptox.epa.gov/rapidtox>

Senior Software Engineer - Contract

American Kennel Club | Raleigh, NC

January 2018 to July 2018

- Modify existing applications to add features and fix bugs
- Consult with product owners to gather requirements for feature enhancements, provide project status updates , and relay technical issues.
- Configure SpringMVC OCMS web app to run on local environment (Mac and PC)
- Document desktop configuration of OCMS web app.
- Web application design, programming and development
- Document software defects, using a bug tracking system
- Fix software defects (bug fix) as reported by users and testers.
- Investigate customer problems referred by technical support.

Contracted project

American Kennel Club OCMS Competition Management System

Add enhancements to the Online Competition Management System (OCMS).

Project technologies: HTML, CSS, JavaScript, jQuery, AngularJS, JavascriptMVC, Java, Spring MVC, SpringSource ToolSuite, Oracle 10, Crucible, JIRA, Jenkins, Velocity Templates, Tomcat Server, Windows 7.

Senior Software Engineer

PeopleFluent | Raleigh, NC

October 2015 to January 2018

- Primary responsibilities include UI Design and implementation for Next Gen PeopleFluent SAS application.
- Implement responsive design features into Next Gen PeopleFluent Application.
- Refactor and rewrite current applications to improve cross browser printing functionality.
- Modify existing applications to add features and fix bugs.
- Refactor current applications enhance feature functionality, improve cross browser compatibility, and appearance.
- Collaborate with product owners, customers or other departments to gather requirements for, provide status reports regarding new applications, and feature enhancements.
- Research and evaluate new software and tools
- Prepare reports or correspondence and documentation concerning project specifications, and status.
- Implemented Bootstrap in the Next Gen framework to provide a more responsive layout wherever possible.
- Involved in High-level and Low-level design discussions of front-end UI architecture.
- Worked with .NET MVC Framework to access data and business logic and build a JavaScript UI to interact with the backend functions.
- Used JSON data in models for two way data binding to update views and communicate with server side methods.
- Created application page templates in MS Visual Studio using C# and .net MVC.
- Assisted QA with pair testing and in developing Test Plans and Test Cases.

Environment: HTML, CSS, JavaScript, jQuery, AngularJS, C# .NET MVC, MS Visual Studio 2015 and 2017 , MS SQL Server, Bamboo, GIT, Windows 7

Senior Software Engineer

American Kennel Club | Raleigh, NC

June 2001 to 2015

- Modify existing applications to add features and fix bugs.
- Research and evaluate new software and tools and refactor or rewrite current applications to adapt them to new technology stack.
- Collaborate with product owners to gather requirements for new applications and feature enhancements
- Mentor junior developers on technology stack and domain knowledge.
- Web application design, programming and development
- Develop data models and databases.
- Prepare reports or correspondence and documentation concerning project specifications, and status.
- Document and resolve software defects, using a bug tracking system
- UI unit testing in Selenium and QUnit.
- Involved in High-level and Low-level design discussions of front-end UI architecture.
- Migrated existing CanJS UI front-end to AngularJS over a year.
- Developed data intensive web application UI for OCMS using AngularJS Models (Services/Factories), Controllers and custom Directives for templates.
- Created application page templates in Spring MVC Framework using JSP, JSTL and Expression Language.
- Followed software development coding best practices.
- Assisted QA with pair testing and in developing Test Plans and Test Cases.
- Work with a team of 5-7 developers and system administrators to convert the AKC Online Registration Application from ColdFusion to a JAVA application using Spring Webflow
- Implement Selenium End to End functional testing using JAVA.
- UI Lead and primary developer for the AKC Breeds application utilizing jQuery, ColdFusion ORM and Hibernate.

Environment: HTML, CSS, JavaScript, jQuery, AngularJS, Java, Spring MVC, SpringSource ToolSuite, Oracle DB, MongoDB, Crucible, JIRA, Jenkins, Velocity Templates, Tomcat Server, Windows 7

Software Engineer

Technology Planning and Management Corporation | Raleigh-Durham, NC

March 2000 to June 2001

- Analyze problems to develop solutions involving computer hardware and software.
- Assign or schedule tasks in order to meet work priorities and goals.
- Evaluate project plans and proposals to assess feasibility issues.
- Consult with users, management, vendors, and technicians to determine computing needs and system requirements.
- Develop new software applications or customize existing applications to meet specific project needs.
- Develop data models and databases.
- Create or modify web-based tools.

Software Engineer

NC Department of Health and Human Services

May 1998 to March 2000

- Analyze problems to develop solutions involving computer hardware and
- software.
- Build online reports of health related data using HTML, ColdFusion, Javascript
- and CSS.
- Manage web server software and ColdFusion server software.

Education

Computer Aided Drafting and Design (Certificate)

Holyoke Community College | Holyoke, MA

1995 to 1997

Skills

Microsoft SQL Server CSS Web Services NuxtJS CI/CD jQuery HTML Bootstrap Vue.js
Cypress Agile XML IT JavaScript GitHub REST Software development Ag-Grid Jira
AngularJS SVN MySQL User Interface (UI) APIs

Additional Information

Authorized to work in the US for any employer